

IR Triage Report

Front Matter

ANALYST NAME: Chibuike Ojukwu

DATE: 2025-09-05

SOURCE: AWS CloudTrail via SIEM

ALERT ID: UR-AWS-20180820-0038

SEVERITY: MEDIUM

DISPOSITION: Escalated

(Escalated to Tier 2. **Attribution:** None at this stage (insufficient evidence to link to a specific threat group)).

Initial Findings

- Incident description:** An IAM user executed an unusually high volume of AWS API calls, including actions against IAM and EC2 services, deviating from its established baseline.
- Scope:** Activity was limited to a single IAM user account in the AWS environment. No evidence of lateral movement or spread was observed.
- Disposition justification:** Escalated due to indicators consistent with possible credential compromise (T1078 – Valid Accounts) and potential IAM manipulation attempts (T1098). The burst of API calls was outside the user’s baseline and targeted high-risk services (IAM/EC2), which raises the likelihood of privilege changes, unauthorized key creation, or rapid resource abuse (e.g., rogue EC2). Further log review and key rotation are required.
- Severity justification:** Medium severity, as the event suggests suspicious behavior that could indicate compromise, though no direct damage or data loss has been confirmed.

Incident Summary

Confirmed Time Bounds (UTC)

- Earliest confirmed event: 2018-08-20T15:15:04Z
- Latest confirmed event: 2018-08-20T15:15:04Z
- Total duration: single timestamp; likely <1 minute.

Timeline of Key Events (UTC)

- 2018-08-20T15:15:04Z** – SIEM detected a sudden burst of API activity from an IAM user.
- 2018-08-20T15:15:04Z** – Behavioral analytics confirmed activity outside baseline involving IAM and EC2 calls.

Scope – Assets & Accounts

Category	Identifier	Role / Business Context	Status
Account	IAM user (redacted)	AWS user account with IAM/EC2 access	At Risk
Service	AWS IAM	Identity and access management	Targeted
Service	AWS EC2	Compute provisioning	Targeted

Impact

- No confirmed impact. No evidence of service disruption, privilege escalation, or data exfiltration. The risk is elevated due to the sensitive nature of IAM and EC2 operations.

MITRE ATT&CK TTPs Observed

- **T1078: Valid Accounts** – Possible misuse of legitimate IAM credentials.
- **T1098: Account Manipulation** – Risk of attempted changes to IAM configurations.
- **T1069: Permission Groups Discovery** – Likely reconnaissance via IAM API activity.

Threat Intelligence Indicators

- **Network:** None identified.
- **Host/Files:** None identified.
- **UA/Tools:** No malicious tools confirmed; activity flagged due to API frequency anomaly.

Recommended Actions

1) IR-Related Actions for Impacted Infrastructure

- Suspend or disable the affected IAM user account pending full investigation.
- Escalate to Tier 2/IR for deep review of CloudTrail logs and IAM activity history.

2) Prevention – Impacted Infrastructure

- Enforce MFA across all IAM users.
- Apply least privilege principle by reviewing and restricting IAM permissions.
- Implement service control policies to prevent excessive or abnormal API activity.

3) Prevention – Internal Policies

- Strengthen IAM governance policies requiring justification for elevated or bulk API operations.
- Provide targeted training for administrators on secure AWS IAM practices.

4) Detection – SOC Flow Improvements

- Refine anomaly detection thresholds for IAM/EC2 API calls to reduce false positives while maintaining sensitivity.
- Correlate IAM anomalies with other AWS service logs for broader context.
- Update SOC playbooks with procedures for AWS IAM anomaly triage.